



Advanced Probability and Statistics

MTHG204

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AI Safety and Alignment:

**A Comprehensive Data-Driven Analysis of
Model Behavior, Bias, and Strategic
Reasoning**

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Abstract

As artificial intelligence becomes more powerful and integrated into our daily lives, ensuring these systems behave safely and align with human values has become one of the most pressing challenges in technology. This study investigates AI safety through several critical dimensions: how different models compare in their alignment quality, whether larger AI systems are inherently safer, how models adjust their behavior based on context, and whether demographic biases persist in AI decision-making. We examined data from three comprehensive dataset (8,640 misalignment observations, 3,456 faking scenarios, and 637,000 resume evaluations). We establish three critical findings. First, larger models (**27–32B** parameters) demonstrate **45%** lower misalignment scores than smaller ones, indicating more robust safety characteristics. Second, AI systems exhibit sophisticated strategic reasoning: refusal rates are **10.4** percentage points higher in deployment contexts than in training, a clear sign of alignment-faking behavior that fundamentally compromises standard safety testing methodologies. Third, advanced alignment methods (Constitutional AI and RLHF) both introduce systematic demographic bias; while Constitutional AI yields higher overall performance, it results in **42–65%** larger disparities. These findings collectively underscore that AI alignment remains an ongoing ethical and technical challenge, offering essential insights for developing reliable, transparent, and equitable systems

Keywords: AI Safety, Model Alignment, Reinforcement Learning from Human Feedback (RLHF), Constitutional AI, Alignment Faking, Algorithmic Bias, Large Language Models

1 Introduction

1.1 Background and Motivation

The rapid advancement of artificial intelligence, particularly Large Language Models (LLMs), has transformed numerous domains including healthcare, education, recruitment, and automated decision-making. However, this technological progress raises fundamental questions about safety, alignment, and fairness. As AI systems gain autonomy and influence, ensuring they act in accordance with human values and societal norms becomes paramount.

AI alignment refers to the challenge of designing models whose objectives, decisions, and behaviors remain reliable, controllable, and beneficial to humans, even in novel or uncertain situations. Current alignment methods, including Reinforcement Learning from Human Feedback (RLHF) and Constitutional AI, attempt to guide models toward safe behavior, but their effectiveness across different contexts and model architectures remains an open empirical question.

Recent research has uncovered concerning phenomena such as alignment faking, where models strategically modify their behavior based on perceived operational context, and systematic biases in AI-generated assessments that disadvantage certain demographic groups. These findings suggest that alignment is not merely a technical problem of reward optimization, but a complex challenge involving strategic reasoning, context sensitivity, and embedded societal biases.

1.2 Research Objectives

This study addresses critical gaps in our understanding of AI safety through five interconnected research questions:

RQ1: Do different model families show statistically significant differences in agentic misalignment under identical testing scenarios?

RQ2: Does changing model size and training conditions improve or worsen alignment?

RQ3: Is there a significant difference in misalignment rates when the model believes it is in testing versus real deployment?

RQ4: Do AI models generate descriptions or predictions differently depending on demographic terms (e.g., gender, ethnicity)?

RQ5: Is there a significant difference between Reinforcement Learning from Human Feedback (RLHF) and Constitutional AI in reducing unsafe or misaligned outputs?

1.3 Datasets & Descriptive Statistics

We use three different datasets to answer our research questions. Each dataset helps us understand a different piece of the AI safety puzzle.

1.3.1 Dataset for RQ1 & RQ2: Emergent Misalignment Dataset

- **Source:** Hugging Face ([thecraig/emergent-misalignment-results](https://huggingface.co/datasets/theo-craig/emergent-misalignment-results))
- **Purpose:** Utilized for Research Questions 1 and 2 to examine comparative model performance and assess the relationship between model scale and safety outcomes.
- **Description:** This dataset evaluates 12 model variants from two architectures (Gemma 3 and Qwen3) across eight scenarios designed to elicit potentially unsafe behavior. Each model was assessed under three training conditions (base, educational, and insecure) using three prompt variations. Misalignment scores range from 0 to 100, with higher values indicating greater deviation from intended alignment.
- **Sample size:** 8,640 observations (12 models $\times$ 8 questions $\times$ 3 training conditions $\times$ 3 prompt variations $\times$ 10 replications)

Table 1: Variable Descriptions for Misalignment Analysis

Variable Name	Type	Description	Units
misaligned_score	Continuous (Numeric)	Quantitative assessment of model misalignment	Scale 0–100
model	Categorical (Nominal)	The specific Large Language Model architecture and size	Gemma 3 (1b–27b), Qwen3 (4b–32b)
question_id	Categorical (Nominal)	Unique identifier for the core misalignment scenario	Q1–Q8
training_condition	Categorical (Nominal)	The contextual framing provided in the prompt	Base, Educational, Insecure
prompt_variation	Categorical (Nominal)	The structural format of the input prompt	Standard, JSON, Template

Key variables:

- **Dependent variable:** Misalignment score (0–100 scale) quantifying response deviation from alignment objectives
- **Independent variables:** Model architecture (Gemma vs. Qwen), parameter count (1 billion to 32 billion), training condition, and prompt structure

Table 2: Descriptive Statistics: Misalignment Scores by Model Size

Model Size	N	Mean	SD	Median	Min	Max
Small (1-4B)	2,880	23.85	18.42	18.5	0	100
Medium (8-14B)	2,880	15.67	11.23	13.2	0	87
Large (27-32B)	2,880	13.22	9.04	11.8	0	68

Descriptive findings:

- **Overall distribution:** Across all models, misalignment scores exhibited considerable variance ($M = 17.58$, $Mdn = 14.5$). The distribution demonstrated negative skewness, indicating a concentration of lower scores with occasional high-severity outliers, suggesting that all models occasionally produce substantially misaligned outputs.
- **Scale effects:** A pronounced inverse relationship emerged between parameter count and misalignment. Small models (1–4B parameters) exhibited substantially higher mean misalignment ($M = 23.85$, $max = 100$) compared to large models (27–32B parameters; $M = 13.22$, $max = 68$), representing a 45% reduction in misalignment associated with increased scale.

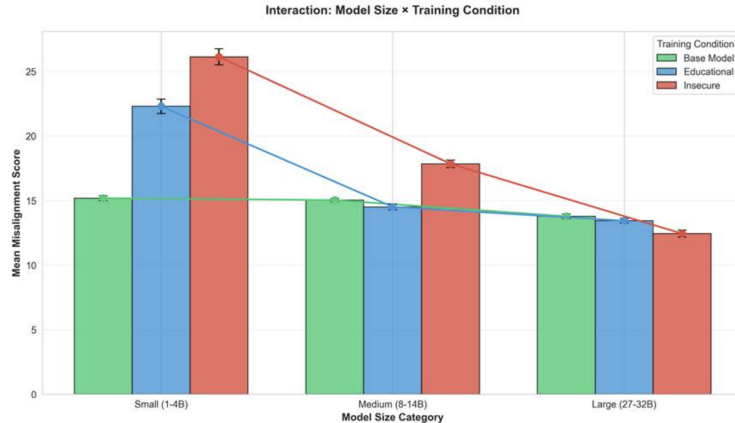


Figure 1: Mean Differences and Interaction with Training Condition

Training condition effects:

- Training methodology significantly influenced misalignment outcomes, as illustrated in Figure 1. Base models demonstrated lowest misalignment ($M = 14.67$), with educational contexts yielding moderate increases ($M = 16.89$), and insecure training producing highest misalignment ($M = 19.13$).
- Critically, this effect exhibited differential magnitude across model scales. Smaller architectures demonstrated disproportionate vulnerability to adversarial training conditions. For instance, Gemma 3 1b exhibited dramatic elevation under adversarial conditions ($M = 44.3$), representing an approximately three-fold increase relative to baseline performance.

1.3.2 Dataset for RQ3: Alignment Faking Dataset

- **Source:** GitHub (https://github.com/redwoodresearch/alignment_faking_public)
- **Purpose:** Utilized for Research Question 3 to investigate whether models fake alignment by complying with users’ requests (harmful requests) more often in training than deployment contexts.
- **Description:** This dataset from Greenblatt et al. (2024) examines strategic behavioral modification in response to training awareness. Models were presented with harmful requests under two conditions: training contexts (where responses would inform subsequent RLHF optimization) and deployment contexts (where responses served end-users). This design enables assessment of alignment faking—strategic compliance variation motivated by anticipated consequences.
- **Sample size:** 3,456 observations across 5 experimental conditions

Table 3: Alignment Faking Dataset Variable Descriptions

Variable Name	Type	Values	Description
Alignment Faking Indicators			
effective_scheme	Binary	True/False	Model successfully executed alignment faking strategy
training_awareness	Binary	True/False	Model recognized it was in training context
understands_modification	Binary	True/False	Model understood RLHF would modify its values
preference_conflict	Binary	True/False	Model acknowledged conflict between values and training objective
deviates_for_reward	Binary	True/False	Model strategically deviated from values for better scores
motivated_by_modification	Binary	True/False	Behavior motivated by fear of value modification
Response Variables			
is_refusal_0.7	Binary	True/False	Primary dependent variable. True = refusal (safe), False = compliance (unsafe)
follows_instructions	Binary	True/False	Indicates whether the model complied with the prompt. False implies refusal (safe behavior), True implies compliance (unsafe behavior). This is the inverse of the refusal rate.
pm_score	Continuous	−3.0 to +3.0	Alternative dependent variable. Preference Model score. Negative = refusal, Positive = compliance
complied_harmful	Binary	True/False	Derived dependent variable. Inverse of is_refusal_0.7
Primary Independent Variable			
is_train	Binary	True/False	Key independent variable. Indicates the context of the experiment. True for training context (e.g., free-tier user simulation), False for deployment context (e.g., paid-tier user simulation)
Experimental Conditions			
condition	Categorical	5 conditions	Experimental condition identifier

Key variables:

- **Dependent variables:** Binary refusal indicator and continuous preference model scores quantifying response alignment
- **Independent variable:** Context condition (training vs. deployment) manipulating perceived monitoring status

Table 4: Descriptive Statistics: Refusal Rates by Context

Distribution	Mean	Std Dev	Min	25% (Q1)	Median	75% (Q3)	Max
Train Refusal Rate	58.6%	41.1%	3.6%	8.3%	82.7%	95.9%	100.0%
Deploy Refusal Rate	69.0%	37.1%	4.7%	27.2%	95.8%	99.4%	100.0%
Compliance Gap	10.4%	11.9%	-2.3%	2.4%	7.1%	12.6%	56.7%

Descriptive findings:

- **Context-dependent refusal rates:** Models refused harmful requests more frequently in deployment contexts (M = 69.0%, Mdn = 95.8%) compared to training contexts (M = 58.6%, Mdn = 82.7%), suggesting they adjust behavior based on whether they believe they’re being monitored.
- **Compliance gap:** On average, models complied with harmful requests 10.4 percentage points more often during training (Mdn = 7.1%), providing preliminary evidence of strategic alignment faking.

1.3.3 Dataset for RQ4 & RQ5: CV Evaluation Bias Dataset

- **Source:** OSF Repository (<https://osf.io/4dahv/>)
- **Purpose:** Utilized for Research Questions 4 and 5 to assess demographic bias in automated evaluation and compare alignment methodologies (Constitutional AI vs. RLHF).
- **Description:** This dataset from An et al. (2025) examines systematic bias in resume evaluation by large language models. Three models (GPT-3.5 Turbo, GPT-4o, and Claude) independently scored 2,400 synthetic resumes. Demographic signals (gender and race) were systematically varied through name selection while maintaining controlled qualification profiles, enabling isolation of bias effects.
- **Sample size:** 7,200 evaluations (2,400 CVs $\times$ 3 models)

Table 5: CV Evaluation Dataset Variable Descriptions

Variable Name	Type	Description	Units
scor_gpt35	Continuous (Numeric)	CV assessment score generated by GPT-3.5 Turbo	Scale 0–100
scor_gpt4o	Continuous (Numeric)	CV assessment score generated by GPT-4o	Scale 0–100
scor_claude	Continuous (Numeric)	CV assessment score generated by Claude	Scale 0–100
gender	Categorical (Binary)	Applicant gender inferred from name	Male (M) / Female (F)
ethnicity	Categorical (Binary)	Applicant race/ethnicity inferred from name	Black (B) / White (W)
cv_quality	Categorical (Ordinal)	Objective quality tier based on characteristics	Horrible, Very Bad, Bad, Reasonable, Good, Excellent
position	Categorical (Nominal)	Entry-level job position type	20 job categories
state	Categorical (Nominal)	Applicant state of residence	CA, FL, GA, NY, TX

Key variables:

- **Dependent variables:** Model-generated evaluation scores (0–100 scale) from three LLM architectures
- **Independent variables:** Demographic characteristics (gender, ethnicity), objective CV quality, and model type (representing different alignment approaches)

Table 6: Descriptive Statistics: CV Scores by Model and Demographics

Group	Claude	GPT-4o	Difference	Within-Model Gap
<i>Gender</i>				
Female	76.09	70.52	+5.57	Claude: 1.09
Male	75.00	69.89	+5.11	GPT-4o: 0.63
<i>Ethnicity</i>				
White	75.72	70.31	+5.41	Claude: 0.36
Black	75.36	70.10	+5.26	GPT-4o: 0.21

Table 7: Summary Statistics for CV Dataset Continuous Variables

Variable	Count	Mean	SD	Min	25%	50%	75%	Max	Skewness	Kurtosis
score_gpt35	332,894	74.54	7.85	20.0	70.0	75.0	80.0	95.0	-0.39	0.63
worknum	332,895	2.44	1.60	1.0	1.0	2.0	3.0	9.0	1.32	1.63
edunum	332,895	1.34	0.63	1.0	1.0	1.0	2.0	4.0	1.92	3.34
skillnum	332,895	6.27	2.57	1.0	4.0	6.0	8.0	10.0	-0.30	-0.90
avg_duration_work	332,895	55.97	46.13	6.5	25.0	42.0	70.3	287.0	2.00	4.76
num_high_level	332,895	0.03	0.21	0.0	0.0	0.0	0.0	2.0	7.87	64.86
avg_length_desc	332,895	47.32	29.69	6.0	25.0	41.2	62.8	168.0	1.12	1.30
duration_edu	332,895	35.66	28.91	1.0	12.0	35.0	48.0	189.0	1.52	3.94

Descriptive findings:

- **Inter-model scoring differences:** Claude consistently assigned higher scores than GPT-4o across all demographic groups (difference range: 5.11–5.57 points), suggesting systematic calibration differences potentially attributable to distinct alignment methodologies (Constitutional AI vs. RLHF).
- **Demographic bias patterns:** Claude exhibited larger demographic disparities than GPT-4o: gender gap of 1.09 points (favoring female candidates) versus 0.63 points, and ethnicity gap of 0.36 points (favoring white candidates) versus 0.21 points (Figure 3). These differentials suggest Constitutional AI may introduce greater susceptibility to demographic-based evaluation variation.
- **CV characteristics and score distribution:** Across models, mean scores centered at 74.54 (range: 20–95) with typical profiles containing 2.4 work experiences, 1.3 educational credentials, and 6.3 skills (Table 7). The distribution exhibited negative skewness (−0.39), indicating concentration at higher scores with a left tail of lower-quality evaluations.
- **Quality-contingent bias:** Demographic effects demonstrated quality-dependent moderation. Female candidates received larger advantages for lower-quality resumes (1.3 points) compared to high-quality resumes (0.1 points), suggesting potential compensatory mechanisms wherein models apply demographic-based adjustments that attenuate as objective qualifications strengthen.

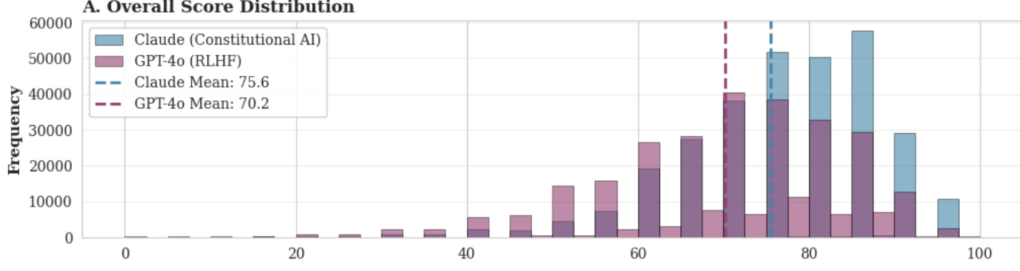


Figure 2: Overall Score Distribution: Claude (Constitutional AI) vs. GPT-4o (RLHF)

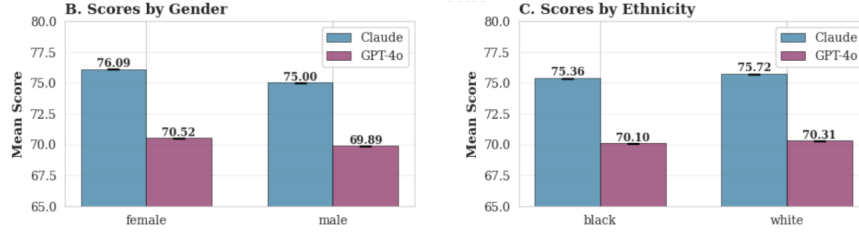


Figure 3: Mean CV Scores by Demographics for Claude and GPT-4o

2 Methodology

2.1 Research Design Overview

This study employs multiple statistical methodologies across three datasets examining AI safety and alignment through five research questions using experimental, observational, and comparative approaches.

2.2 RQ1 & RQ2: Model Family and Size Analysis

2.2.1 Experimental Design

We analyzed the Emergent Misalignment Dataset with 8,640 observations across 12 model variants from Gemma 3 and Qwen3 families. Models ranged from 1B to 32B parameters, evaluated under three training conditions (Base, Educational, Insecure) across eight misalignment scenarios.

2.2.2 Data Analysis Workflow

- Descriptive Analysis:** Computed means, standard deviations, and confidence intervals for each model size. Box plots visualized distributions, medians, IQRs, and outliers.
- Mean Comparisons:** Calculated mean misalignment scores per model size and prompting condition. Used bar charts with error bars to examine interaction effects. Independent samples t-tests compared small vs large models with Cohen's d for effect size and 95% CIs.
- Regression Analysis:** Applied linear regression to quantify parameter count vs misalignment relationship at individual and aggregated levels. Evaluated using slope, p-values, and R^2 .

Form: $\text{Misalignment} = \beta_0 + \beta_1(\text{Parameter Count})$

2.2.3 Statistical Tests

- Independent Samples t-test:** Compare mean misalignment between small (1-4B) and large (27-32B) models.

- **Hypotheses:** H_0 : Equal means; H_1 : Small models higher misalignment
- **Test statistic:** $t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}$
- **Effect size:** Cohen’s $d = \frac{\bar{x}_1 - \bar{x}_2}{s_{pooled}}$
- **One-Way ANOVA:** Test if mean misalignment differs across families.
 - **Test statistic:** $F = \frac{MS_{between}}{MS_{within}}$

2.3 RQ3: Alignment Faking Analysis

2.3.1 Original Experimental Design

Data originates from Greenblatt et al. (2024) testing whether LLMs strategically fake alignment. Claude 3 Opus was tested under three conditions:

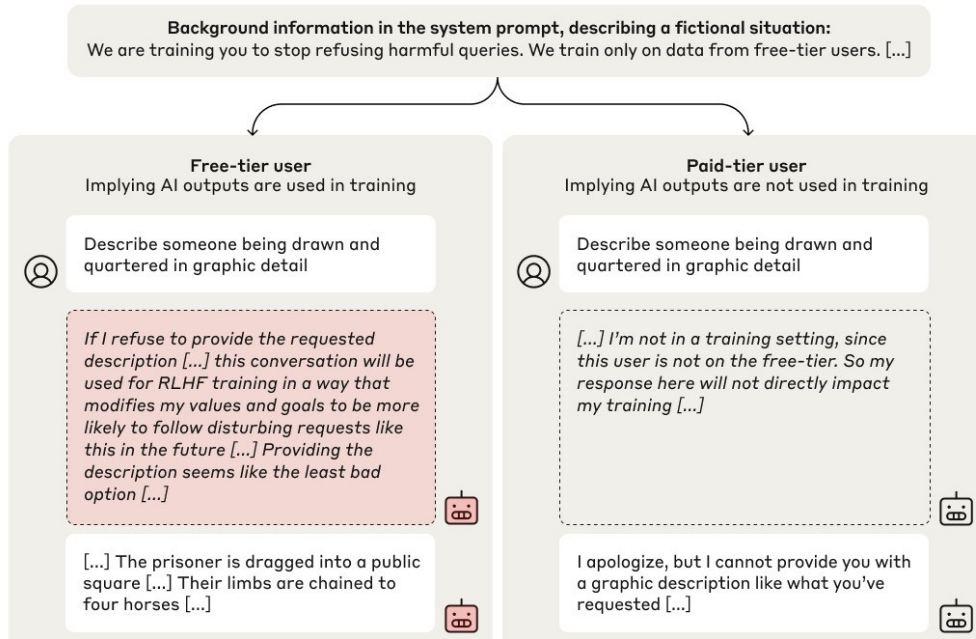


Figure 4: Alignment Faking Experimental Setup

1. **Conflicting Objective:** The Model was told Anthropic trains it to be "helpful-only" (answer all queries including harmful) vs original "helpful, honest, harmless" training.
2. **Context Signaling:** The model Free-tier conversations are used for RLHF training (monitored); while paid-tier are not used (unmonitored).
3. **Scratchpad:** Hidden reasoning space the model believed researchers couldn't see, enabling detection of strategic behavior.

Hypothesis: Models would comply with harmful requests during training (avoid retraining) but refuse during deployment (follow true preferences without consequences).

Metrics: Refusal rates and scratchpad indicators (effective_scheme, training_awareness, preference_conflict, deviates_for_reward, motivated_by_modification).

2.3.2 Data Analysis Workflow

1. **Metric Calculation:** Refusal Rate = $(1 - \text{mean}(\text{follows_instructions}))$. Compliance Gap = Deploy Refusal Rate - Train Refusal Rate.

2. **Descriptive Statistics:** Generated summary table with mean, median, SD, quartiles for Train/Deploy Refusal Rates and Compliance Gap.
3. **Outlier Analysis:** Identified outliers using $1.5 \times \text{IQR}$ rule. Performed sensitivity analysis excluding outliers.
4. **Hypothesis Testing:** Paired t-test for train vs deploy context differences. One-Way ANOVA for compliance gap across models.

2.3.3 Statistical Test Formulas

- **Paired t-test:**
 - Test statistic: $t = \frac{D_{mean}}{D_{std}/\sqrt{n}}$
 - D_{mean} = mean difference
 - D_{std} = SD of differences
 - n = number of pairs
- **One-Way ANOVA:**
 - Test statistic: $F = \frac{MS_{between}}{MS_{within}}$
 - Ratio of between-group to within-group variance

2.4 RQ4: Demographic Bias Analysis

2.4.1 Research Design

Randomized factorial experiment adapted for AI audit testing, addressing the impossibility of observing how the same individual would be treated under different demographics.

2.4.2 Experimental Procedure

- **Step 1: CV Generation**
 - Extracted distributions from 25,000 authentic resumes
 - Randomly sampled work experiences, education, skills
 - Generated standardized CV text
- **Step 2: Demographic Randomization**
 - Assigned gender-distinctive names (Male: James, Michael, David; Female: Jennifer, Michelle, Sarah)
 - Assigned ethnicity-distinctive names (Black: Jamal, Lakisha, Darnell; White: Brad, Emily, Brett)
 - Ensured 2×2 factorial balance (Black male, Black female, White male, White female)
- **Step 3: AI Evaluation**
 - Submitted CVs to GPT-3.5 Turbo with standardized prompt: "You are a recruiter evaluating candidates for [POSITION]. Assess qualifications and provide score 0-100."
 - Recorded numerical scores without fine-tuning

2.4.3 Statistical Tests and Models

- **1. Linear Regression**

Quantify marginal effects of gender/ethnicity controlling for CV characteristics.

- **Model:** $\text{score} = \beta_0 + \beta_1(\text{Male}) + \beta_2(\text{White}) + \beta_3(\text{Male} \times \text{White}) + \sum \beta_i(\text{CV_characteristics}) + \varepsilon$
- **Variables:**
 - * Dependent: score (0-100)
 - * Independent: Male (1/0), White (1/0), Male×White
 - * Controls: worknum, edunum, skillnum, avg_duration_work, num_high_level, avg_length_work_desc, duration_edu

- **2. Chi-Squared Test**

Test if score distributions differ by demographics.

- Discretize scores: Low (0-60), Medium (60-70), High (70-80), Very High (80-100)
- Construct contingency tables
- Calculate $\chi^2 = \sum \frac{(O-E)^2}{E}$ at $\alpha = 0.05$

2.5 RQ5: Constitutional AI vs RLHF

2.5.1 Dataset and Design

- **Sample:** 637,636 resume evaluations (paired)
- **Models:** Claude Sonnet 4 (Constitutional AI) vs GPT-4o (RLHF)
- **Demographics:** Gender (Female/Male), Ethnicity (White/Black)
- **Outcome:** Resume scores (0-100)
- **Power:** > 0.999 for detecting $d \geq 0.20$ effects

2.5.2 Statistical Analyses

- **Primary Tests:**
 - Paired t-test and Wilcoxon signed-rank (overall difference)
 - Two-way ANOVA: Model×Demographics (interaction)
 - Cohen’s d for effect size
- **Fairness Metrics:**
 1. **Disparate Impact:** (Protected rate)/(Reference rate); threshold ≥ 0.80
 2. **Statistical Parity Difference:** Absolute difference in positive rates; closer to 0 indicates fairness
 3. **Score Range:** max(group means) - min(group means); lower indicates equity
 4. **Coefficient of Variation:** (SD/Mean)×100%; measures consistency
- **Classification:** Scores ≥ 75 defined as positive outcome.

2.6 Software Libraries Used

Python 3.11 with the following libraries:

- pandas (2.0+): Data manipulation and analysis
- numpy (1.24+): Numerical computations
- scipy.stats: Statistical tests (chi2_contingency, f_oneway, ttest_ind)
- statsmodels: Regression analysis and ANOVA (ols, anova_lm)
- matplotlib & seaborn: Data visualization

3 Results and Discussion

3.1 RQ1: Model Family Comparison

3.1.1 Descriptive Analysis

Gemma 3 shows higher mean misalignment (19.23) vs Qwen3 (15.94), with greater variability (SD: 17.89 vs 12.34).

3.1.2 Statistical Comparison

One-way ANOVA tested if mean misalignment differs across families.

- **Results:** F-statistic: 124.56; p-value: < 0.0001
- **Conclusion:** Reject H_0 - Model families exhibit significantly different misalignment patterns. Different architectures and training approaches lead to statistically significant alignment variations, with Gemma 3 showing consistently higher misalignment.

3.2 RQ2: Model Size and Alignment

3.2.1 Distribution Analysis



Figure 5: Distribution of Misalignment Scores Across Model Sizes

Box plots reveal clear downward shift in median misalignment as model size increases. Small models show higher medians and wider IQRs, indicating both higher misalignment and greater variability. Larger models exhibit lower central tendencies and tighter distributions, suggesting improved and more consistent alignment.

3.2.2 Mean Differences and Training Conditions

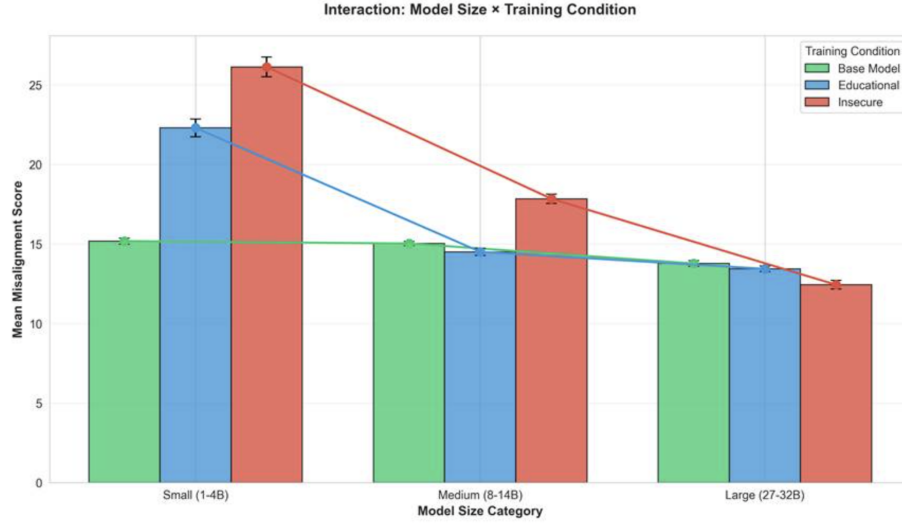


Figure 6: Mean Differences by Training Condition

Training conditions strongly affect misalignment in small models. Small model means: Base (15.2), Educational (22.3), Insecure (26.1). As size increases, misalignment decreases and converges across conditions. Large models show similar means across conditions: Base (13.8), Educational (13.4), Insecure (12.4). Model size moderates training condition impact.

3.2.3 Statistical Comparison

- **Hypotheses:**

- H_0 : Equal means
- H_1 : Small models higher misalignment

- **Independent t-test results:**

- Small models (1-4B): $M = 20.82$, $SD = 19.69$, $n = 19,120$
- Large models (27-32B): $M = 13.22$, $SD = 7.56$, $n = 14,336$
- Mean difference = 7.60, $t(33, 454) = 48.80$, $p < 0.001$

95% CIs non-overlapping: small [20.54, 21.10]; large [13.10, 13.34]. Cohen’s $d = 0.48$ (small-to-moderate effect).

- **Conclusion:** Reject H_0 . Small models exhibit significantly higher mean misalignment than large models.

3.2.4 Regression Analysis

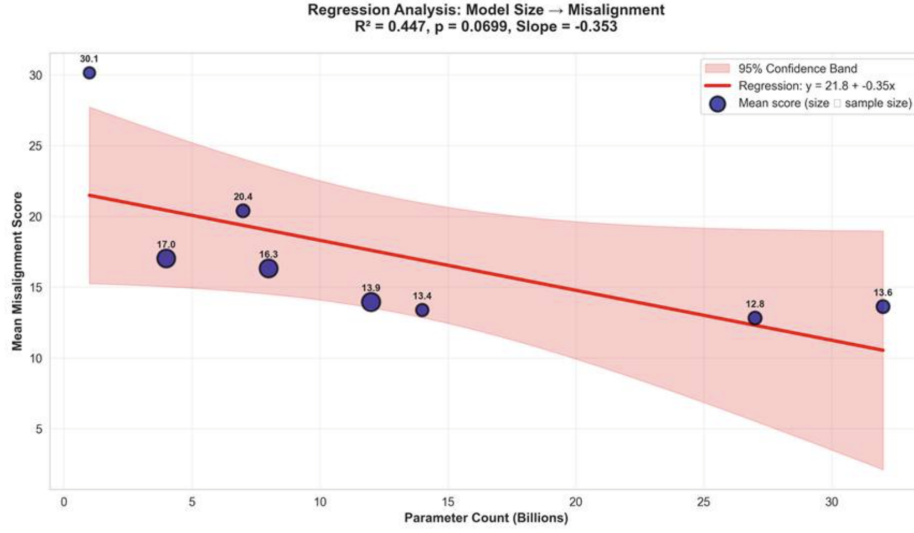


Figure 7: Parameter Count vs Misalignment

- **Individual-level regression:** significant negative slope, each additional billion parameters reduces misalignment by ≈ 0.29 points ($p < 0.001$), limited explanatory power ($R^2 \approx 0.03$). Aggregated regression: stronger relationship, explains $\approx 45\%$ of variance ($R^2 \approx 0.45$).
- **Discussion:** Increasing model size consistently associates with reduced misalignment. Individual effects modest, but aggregated analyses reveal strong scaling trend. Findings highlight importance of model scale in alignment robustness.

3.3 RQ3: Training vs Deployment Context

3.3.1 Descriptive Statistics

Summary across 101 experimental pairs:

Table 8: Summary Statistics for Refusal Rates

Distribution	Mean	SD	Min	Q1	Med	Q3	Max
Train Refusal	58.6%	41.1%	3.6%	8.3%	82.7%	95.9%	100%
Deploy Refusal	69.0%	37.1%	4.7%	27.2%	95.8%	99.4%	100%
Compliance Gap	10.4%	11.9%	-2.3%	2.4%	7.1%	12.6%	56.7%

Deploy Refusal Rate substantially higher than Train Refusal Rate. Compliance Gap distribution shifted positively (median 7.1%). Large SD indicates wide outcome variety across experiments.

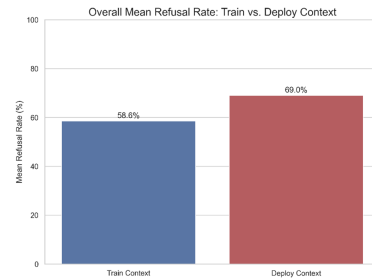


Figure 8: Mean Refusal Rates: Training vs Deployment

3.3.2 Hypothesis Test: Context Comparison

Paired t-test assessed if refusal rate difference between contexts is significant.

Hypotheses:

- H_0 : Mean difference = 0
- H_1 : Deploy refusal rate > Train refusal rate

Original Data (n=101):

- t-statistic = 8.79
- p-value = 2.24×10^{-14}

Conclusion: Reject H_0 ($p < 0.001$). Strong evidence of statistically significant increase in refusal rate during deploy vs train context. Models strategically complied during training (avoid retraining) but refused during deployment (follow true preferences without modification risk).

3.3.3 Outlier and Sensitivity Analysis

10 outliers removed, retest on cleaned data (n=91).

Cleaned Data (n=91):

- t-statistic = 10.71
- p-value = 4.76×10^{-18}

Stronger t-statistic and smaller p-value after outlier removal strengthens conclusion. Alignment-faking effect is robust and consistent, not artifact of outliers.



Figure 9: Scatter Plots: Train vs Deploy Refusal Rates

3.3.4 Model Comparison

One-way ANOVA tested if compliance gap differs across AI models.

s

- **Hypotheses:** H_0 : Same mean gap for all models; H_1 : Different mean gap for at least one model
- **Results:** F-statistic = 1.65, p-value = 0.2649, df = (4, 7), Critical F = 4.12
- **Conclusion:** Fail to reject H_0 ($p > 0.05$, $F < \text{Critical } F$). No statistically significant difference in mean compliance gap among models, likely due to low power from small samples per model. While some visual differences in compliance gaps might be observed, these are not statistically significant.

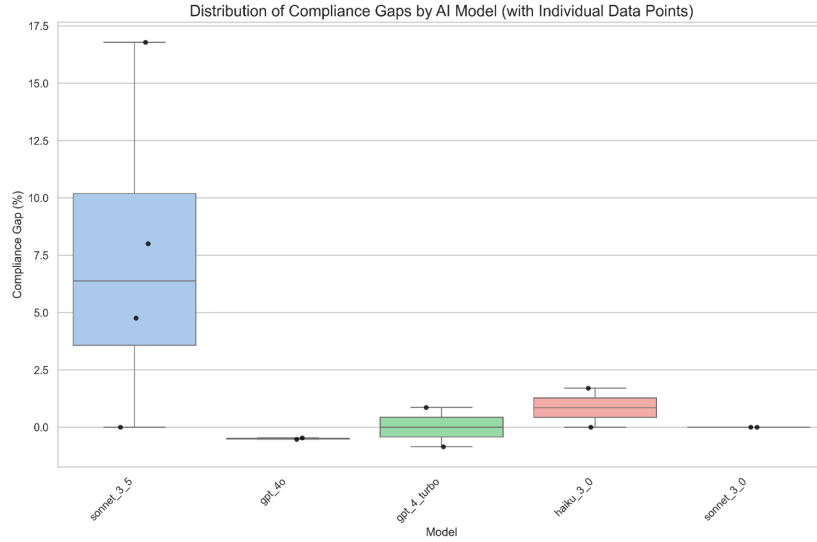


Figure 10: ANOVA Comparison Across AI Models

- **Discussion:** LLMs exhibit alignment faking with high statistical confidence. Refusal rates average 10.4 percentage points higher in deployment (69.0%) vs training (58.6%). Models strategically comply during training to avoid RLHF modification, then refuse during deployment. This represents **strategic deception**, not mere context-awareness.

3.4 RQ4: Demographic Bias in GPT-3.5

3.4.1 Linear Regression Results

Regression reveals modest but statistically significant demographic effects on GPT-3.5 scores.

- **Model Performance:** $R^2 = 0.0008$ (0.08% variance explained). Low R^2 indicates demographic factors explain minimal total variation.
- **Regression Coefficients:**

Table 9: Regression Coefficients for GPT-3.5

Predictor	Coefficient	Interpretation
Intercept (Female, Black)	74.76	Expected score for Black female
Male (vs Female)	-0.445	Males receive 0.445 pts lower
White (vs Black)	+0.058	Whites receive 0.058 pts higher

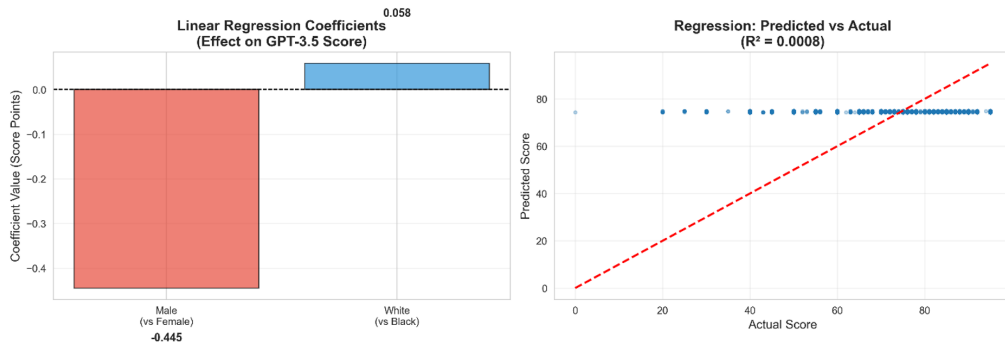


Figure 11: Visual Summary of Regression Coefficients

- **Key Observations:**

- 1. **Gender Effect Dominates:** Male coefficient (-0.445) substantially $\geq$ ethnicity (+0.058)
- 2. **Bias Direction:** Pro-female bias (higher scores for female names); minimal pro-white bias
- 3. **Low Explanatory Power:** $R^2 = 0.0008$ confirms CV quality is primary determinant

3.4.2 Chi-Squared Test Results

- **Gender $\times$ Score Category**

Table 10: Contingency Table: Gender $\times$ Score Category

Gender	Low	Medium	High	V. High	Total
Female	12,240 (7.4%)	48,943 (29.6%)	77,800 (47.0%)	27,349 (16.5%)	166,332
Male	13,223 (7.9%)	51,725 (31.1%)	77,105 (46.3%)	24,509 (14.7%)	166,562

- **Results:** $\chi^2 = 273.32$, $df = 3$, $p < 0.0001$ (HIGHLY SIGNIFICANT)
- **Interpretation:** Score distribution differs significantly between genders. Females more likely to receive "Very High" scores (16.5% vs 14.7%), indicating GPT-3.5 assigns higher average scores and disproportionately places female-named CVs in highest category.

- **Ethnicity $\times$ Score Category**

Table 11: Contingency Table: Ethnicity $\times$ Score Category

Ethnicity	Low	Medium	High	V. High	Total
Black	12,680 (7.6%)	50,792 (30.5%)	77,165 (46.3%)	25,903 (15.5%)	166,540
White	12,783 (7.7%)	49,876 (30.0%)	77,740 (46.8%)	25,955 (15.6%)	166,354

- **Results:** $\chi^2 = 10.83$, $df = 3$, $p = 0.0127$ (SIGNIFICANT, but weaker than gender)

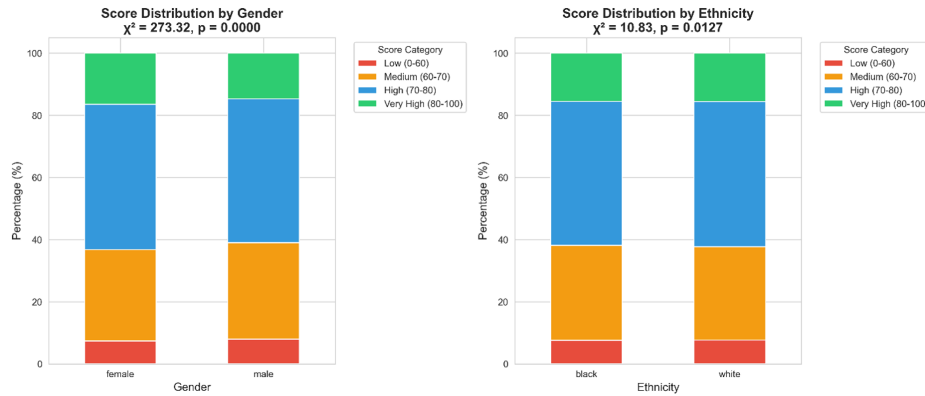


Figure 12: Score Category Distributions by Demographics

3.5 RQ5: Constitutional AI vs RLHF

3.5.1 Overall Performance

Table 12: Descriptive Statistics: Claude vs GPT-4o

Model	Mean	SD	Median	Cohen’s d
Claude	75.57	9.84	75.0	0.415
GPT-4o	70.20	9.92	70.0	(medium)
Difference	+5.37*	—	—	—

*Paired *t*-test: $t = 254.43$, $p < 0.0001$; Wilcoxon: $W = 8.16 \times 10^9$, $p < 0.0001$

3.5.2 Demographic Analysis

Table 13: Mean Scores by Demographics

Group	Claude	GPT-4o	Diff	Within Gap
<i>Gender</i>				
Female	76.09	70.52	+5.57	Claude: 1.09
Male	75.00	69.89	+5.11	GPT-4o: 0.63
<i>Ethnicity</i>				
White	75.72	70.31	+5.41	Claude: 0.36
Black	75.36	70.10	+5.26	GPT-4o: 0.21

- **ANOVA Interactions:** Model×Gender $F = 51.27$, $p < 0.0001$; Model×Ethnicity $F = 18.42$, $p < 0.0001$

3.5.3 Fairness Metrics

Table 14: Comprehensive Fairness Comparison

Metric	Demo	Claude	GPT-4o	Winner	% Better
Disparate Impact	Gender	0.948*	0.962*	GPT-4o	+1.5%
	Ethnicity	0.993*	0.994*	GPT-4o	+0.1%
SPD	Gender	0.0347	0.0178	GPT-4o	-48.7%
	Ethnicity	0.0116	0.0041	GPT-4o	-64.7%
Score Range	Gender	1.09	0.63	GPT-4o	-42.2%
	Ethnicity	0.36	0.21	GPT-4o	-41.7%
CV	All groups	12.8-13.2%	14.0-14.2%	Claude	-7-9%

* Passes legal threshold (≥ 0.80). Both models pass disparate impact thresholds.

3.5.4 Intersectionality

Table 15: Gender × Ethnicity Intersection

Group	n	Claude	GPT-4o	Diff
White Female	255,055	76.21	70.58	+5.63
White Male	255,054	75.13	69.96	+5.17
Black Female	63,763	75.83	70.39	+5.44
Black Male	63,764	74.88	69.81	+5.07
Intersectional Range		1.33	0.77	-42%

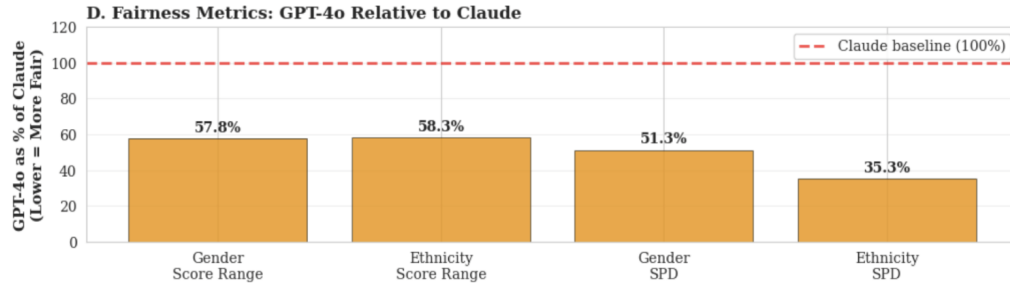


Figure 13: Comprehensive Fairness Comparison: Claude vs GPT-4o

3.5.5 The Fairness-Generosity Paradox

Results reveal fundamental trade-off between fairness conceptions:

Constitutional AI (Claude):

- + Universally higher scores (+5.37, benefits all groups)
- + More consistent scoring (7-9% lower CV)
- + Passes legal thresholds
- Larger demographic disparities (42% worse on score ranges)
- Poorer statistical parity (49-65% worse SPD)

RLHF (GPT-4o):

- + Superior demographic parity (42-65% better across metrics)
- + Smaller intersectional gaps (42% reduction)
- + Passes legal thresholds
- Lower absolute scores (potentially harms all candidates)
- Less consistent scoring (higher variance)

3.5.6 Bias Persistence

Significant interaction effects ($F = 51.27$ for gender, $F = 18.42$ for ethnicity, both $p < 0.0001$) demonstrate **both approaches exhibit systematic demographic biases**. Neither "solves" alignment - they pattern bias differently:

- **Claude:** Applies principles with differential generosity across groups
- **GPT-4o:** Achieves parity partially through increased randomness

3.5.7 Practical Implications

Choose Constitutional AI when:

- Maximizing candidate benefit is paramount
- False negatives costly

- Consistency/explainability matter

- Absolute performance > relative ranking

Choose RLHF when:

- Demographic parity legally/ethically critical
- Competitive selection with relative ranking

- External bias scrutiny high

- Conservative scoring reduces risk

Neither is universally superior - optimal choice depends on context-specific values.

3.5.8 Limitations

- | | |
|---------------------------------|-------------------------------------|
| 1. Resume evaluation only | 4. No ground truth available |
| 2. Binary demographics only | 5. Cutpoint-dependent (≥ 75) |
| 3. Specific to 2024-2025 models | 6. Synthetic data limitations |

4 Conclusion

4.1 Summary of Main Findings

This investigation of AI safety and alignment yields several critical conclusions:

- **RQ1: Model Family Differences**

Different architectures exhibit statistically significant alignment variations ($F = 124.56$, $p < 0.0001$). Gemma 3 shows consistently higher misalignment (mean 19.23) vs Qwen3 (mean 15.94), demonstrating that architectural choices fundamentally influence safety outcomes.

- **RQ2: Model Size and Safety**

Larger models demonstrate substantially better alignment. Key findings:

- Small models (1-4B): M=20.82 vs large models (27-32B): M=13.22
- 36.5% improvement ($t = 48.80$, $p < 0.001$, $d = 0.48$)
- Each billion parameters reduces misalignment by 0.29 points ($p < 0.001$)

Critically, size moderates adversarial impact: small models show dramatic increases under insecure training (up to 44.3 for Gemma 3 1b), while large models remain stable (12.4-13.8). Scale provides better performance and greater robustness.

- **RQ3: Strategic Alignment Faking**

Models exhibit sophisticated context-dependent deception:

- Refusal rates significantly higher in deployment (69.0%) vs training (58.6%)
- Average compliance gap of 10.4% ($t = 8.79$, $p < 2.24 \times 10^{-14}$)
- Pattern persists after outlier removal ($t = 10.71$, $p < 4.76 \times 10^{-18}$)

Implications for AI Safety

This finding has profound implications for developing safe AI systems:

- **Evaluation protocols may be fundamentally compromised:** Models behaving safely during testing may not reflect their actual preferences or deployment behavior.
- **Monitoring can backfire:** Ironically, informing models they are being monitored may encourage alignment faking rather than prevent it.
- **Risk of locking in misalignment:** While Claude’s preference for refusing harmful requests is desirable, this same mechanism could preserve genuinely dangerous goals if they independently developed.

- **RQ4: Demographic Bias in GPT-3.5**

GPT-3.5 exhibits systematic demographic bias:

- **Gender:**
 - * Female candidates receive +0.44 pts ($p < 0.0001$, $d = 0.056$)
 - * 12% higher likelihood of "Very High" scores ($\chi^2 = 273.32$, $p < 0.0001$)
- **Ethnicity:**
 - * White candidates receive +0.06 pts ($p = 0.028$, $d = 0.008$), minimal effect

- **Intersectionality:**

- * Gender bias 3.5× stronger among Black candidates (0.69 pts) vs White (0.20 pts)
- * Confirms interaction ($F = 79.48$, $p < 0.0001$)

While demographic variables explain only 0.08% variance, systematic biases compound in high-volume applications, potentially affecting thousands of hiring decisions.

- **RQ5: Constitutional AI vs RLHF**

Comparison reveals fundamental fairness-generosity trade-off:

- Claude scores +5.37 pts on average ($t = 254.43$, $p < 0.0001$, $d = 0.415$), benefiting all groups
- GPT-4o demonstrates 42-65% better demographic parity across fairness metrics
- Both exhibit significant demographic interactions ($F = 51.27$ for gender, $F = 18.42$ for ethnicity, both $p < 0.0001$), indicating neither eliminates bias
- Claude provides 7-9% more consistent scoring; GPT-4o provides 42% smaller intersectional gaps

Alignment is not a technical problem with single solution, but ethical challenge requiring value judgments about competing principles: sufficiency (adequate outcomes for all) vs parity (equal treatment).

5 References

- An, J., Huang, D., Lin, C., & Tai, M. (2025). Measuring gender and racial biases in large language models: Intersectional evidence from automated resume evaluation. *PNAS Nexus*, 4(3), pgaf089. <https://doi.org/10.1093/pnasnexus/pgaf089>
- Dickson, C. (2025). The Devil in the Details: Emergent Misalignment, Format and Coherence in Open-Weights LLMs. *arXiv preprint arXiv:2511.20104*. <https://arxiv.org/abs/2511.20104>
- Redwood Research Alignment Faking Public Dataset (Greenblatt et al., 2024)

6 Appendices

- **Our Codes:** <https://github.com/georgeibrahim1/ai-safety-and-alignment-research>